USPAS Lecture: More on RF Cavity Types and Design

Patrick Tutt

July 22, 2026

This material is based upon work supported by the U.S. Department of Energy, Office of Science, Office of Nuclear Physics and used resources of the Facility for Rare Isotope Beams (FRIB) Operations, which is a DOE Office of Science User Facility under Award Number DE-SC0023633.

Outline

- A little bit more on the pillbox…
- Solution for low-beta acceleration
- Cavity design/optimization

So you want to build a cavity... (for acceleration)

So you want to build a cavity... (for acceleration)

- Some natural questions:

1. **Geometry: Is the pillbox shape sufficient for any particle types/energies?**

2. **EM: How much field can I generate in the cavity to accelerate the beam?**

3. **Materials: What metals are good choices? Will the cavities be super-conducting or normal-conducting?**

4. **Scaling: How many cavities do I need?**

So you want to build a cavity... (for acceleration)

- Some natural questions:

1. **Geometry: Is the pillbox shape sufficient for any particle types/energies?**
2. **EM: How much field can I generate in the cavity to accelerate the beam?**
3. **Materials: What metals are good choices? Will the cavities be super-conducting or normal-conducting?**
4. **Scaling: How many cavities do I need?**

- Depends on:
  - Application (What beams are being accelerated? To what energy?)
  - Mode of operation (CW or pulsed)

Is the pillbox good enough?

- So far we have only discussed the pillbox → can we use this cavity shape for all applications?
- Remember rest mass of particles:
  - Electron: 0.511 MeV/c^2
  - Proton: 938 MeV/c^2

$$E_{kin} = (\gamma - 1)mc^2 \qquad \gamma = \sqrt{1 - \frac{1}{\beta^2}}$$

- Compare β and ɣ:

Particle Type:	ɣ @ $E_{kin} = 500\ keV$	ɣ @ $E_{kin} = 50\ MeV$	ɣ @ $E_{kin} = 500\ MeV$
Electron	1.978	98.85	979.5
Proton	1.001	1.053	1.533

Particle Type:	β @ $E_{kin} = 500\ keV$	β @ $E_{kin} = 50\ MeV$	β @ $E_{kin} = 500\ MeV$
Electron	0.863	0.9999	0.999999
Proton	0.033	0.314	0.758

Is the pillbox good enough?

- So far we have only discussed the pillbox → can we use this cavity shape for all applications?
- Remember rest mass of particles:
  - Electron: 0.511 MeV/c^2
  - Proton: 938 MeV/c^2

$$E_{kin} = (\gamma - 1)mc^2 \qquad \gamma = \sqrt{1 - \frac{1}{\beta^2}}$$

- Compare β and ɣ:

Particle Type:	ɣ @ $E_{kin} = 500\ keV$	ɣ @ $E_{kin} = 50\ MeV$	ɣ @ $E_{kin} = 500\ MeV$
Electron	1.978	98.85	979.5
Proton	1.001	1.053	1.533

Particle Type:	β @ $E_{kin} = 500\ keV$	β @ $E_{kin} = 50\ MeV$	β @ $E_{kin} = 500\ MeV$
Electron	0.863	0.9999	0.999999
Proton	0.033	0.314	0.758

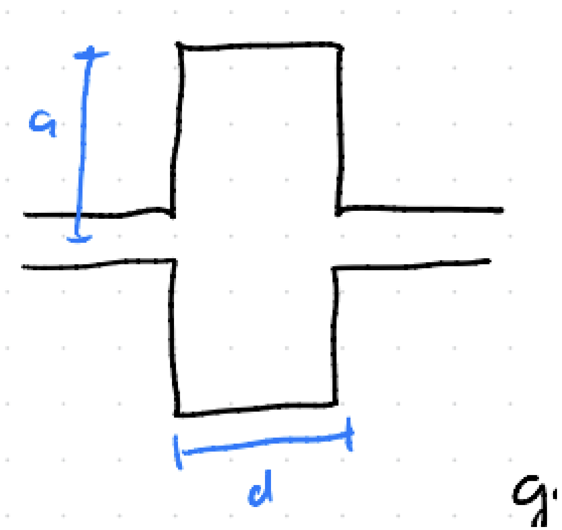
Takeaway: Electrons reach β ≈ 1 after only a few MV and then the velocity is “locked”.
Protons/ions require much more energy to reach higher β’s.

Is the pillbox good enough?

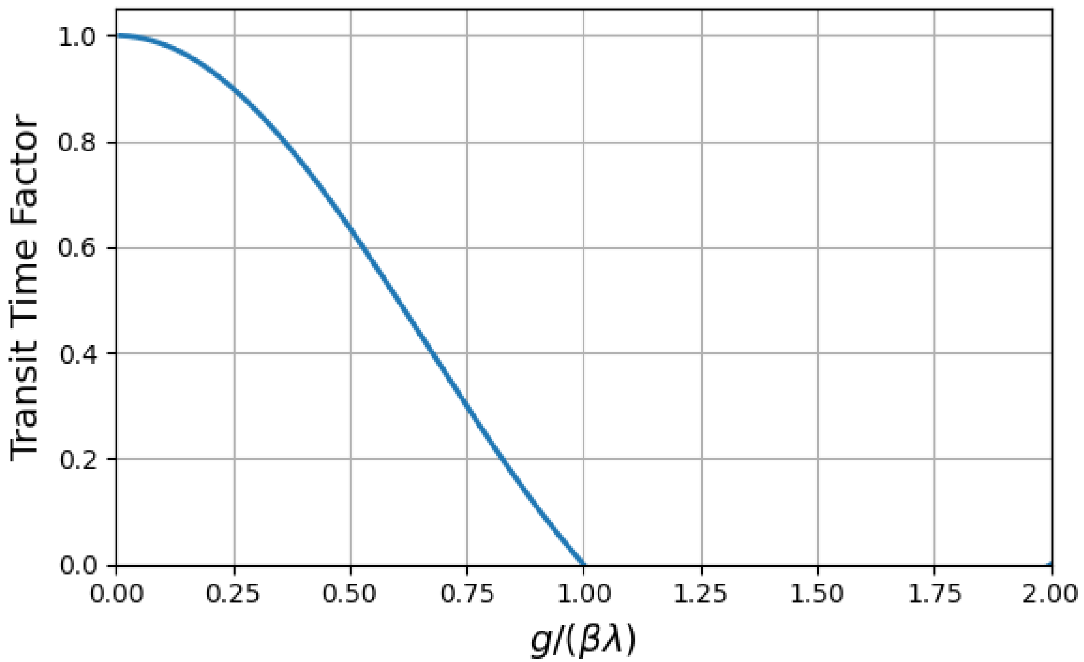
- How can we tell if the pillbox is a good choice for certain $\beta's$ → Transit Time Factor (TTF)
- Remember the transit time factor from Monday's lecture for a pillbox cavity:
  - $\omega = 2\pi f$ (resonant frequency)
  - g = gap distance
  - β = $^{v}/_{c}$

$$T = \frac{\sin\left(^{\omega g}/_{2\beta c}\right)}{^{\omega g}/_{2\beta c}}$$

- Plotting TTF as a function of $^{g}/_{\beta\lambda}$:



Pillbox Drawing

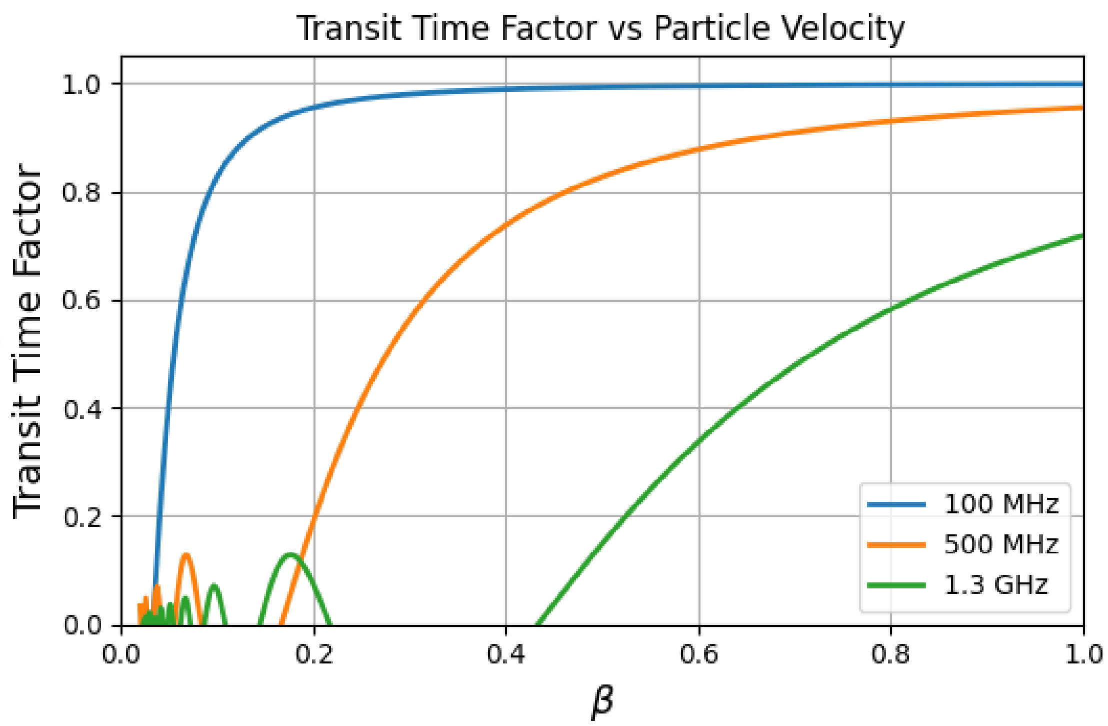


Is the pillbox good enough?

- How can we tell if the pillbox is a good choice for certain $\beta's$ → Transit Time Factor (TTF)
- Remember the transit time factor from Monday's lecture for a pillbox cavity:
  - $\omega = 2\pi f$ (resonant frequency)
  - g = gap distance
  - $\beta = {}^{v}/_{c}$

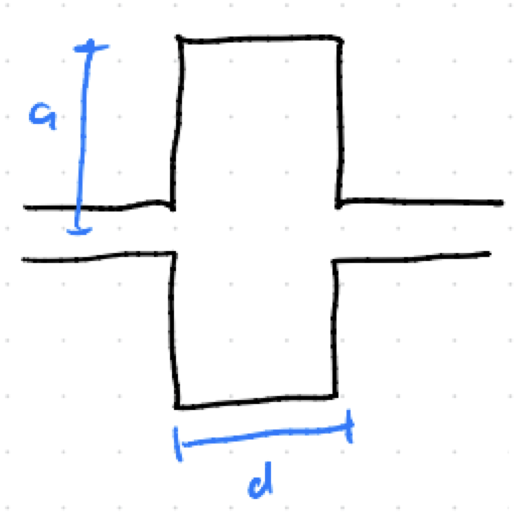
$$T = \frac{\sin\left({}^{\omega g}/_{2\beta c}\right)}{{}^{\omega g}/_{2\beta c}}$$

- Based on previous plot, to increase TTF at low β:
  - Reduce the frequency of the cavity
  - Reduce the gap distance
- Varying the frequency for fixed gap width (10cm)
- The problem:
  - Pillbox fundamental mode: $\omega = \frac{2.405 * c}{2\pi R}$
  - For 100 MHz → R = 1.15m



Is the pillbox good enough?

Pillbox

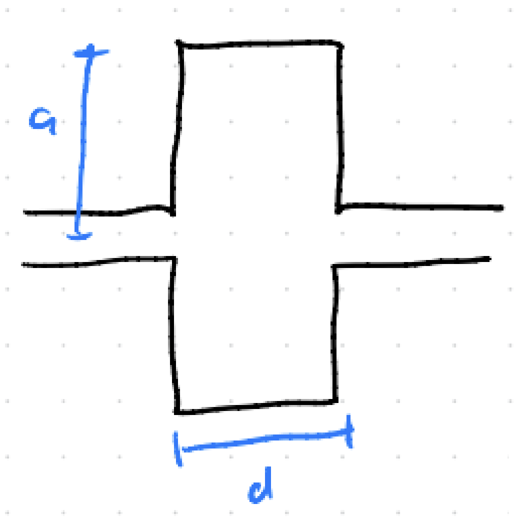


Sufficient for high β → Seems good in principle, but any practical limitations?

Not sufficient for low β → Can't use it. Will need another design

Is the pillbox good enough?

Pillbox

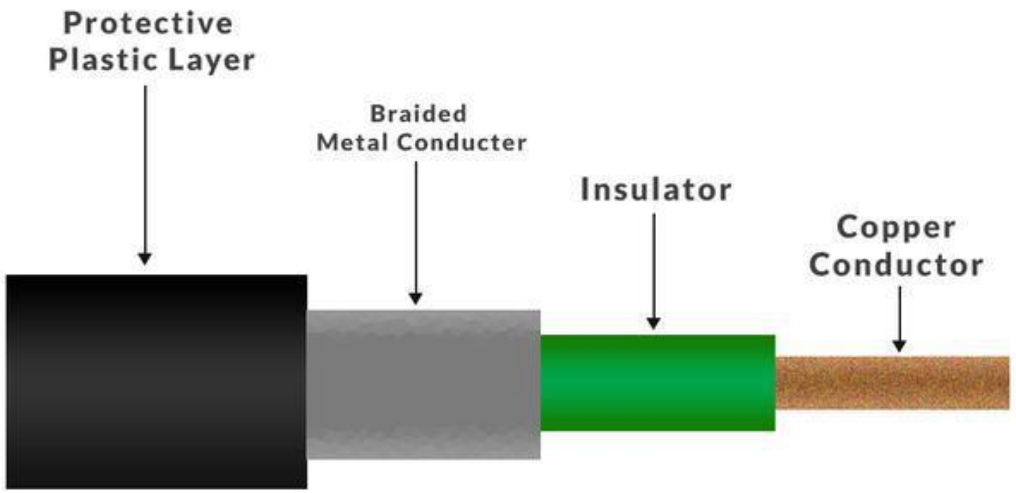


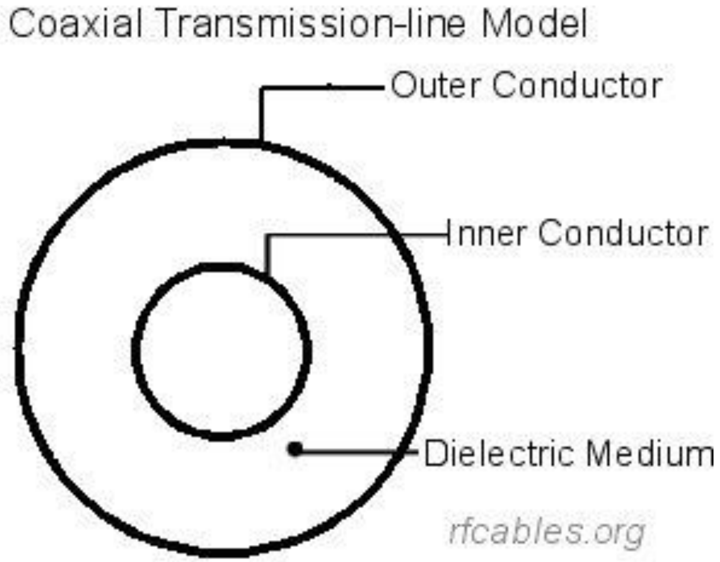
Sufficient for high β → Seems good in principle, but any practical limitations?

Not sufficient for low β → Can't use it. Will need another design

RF Transmission Lines

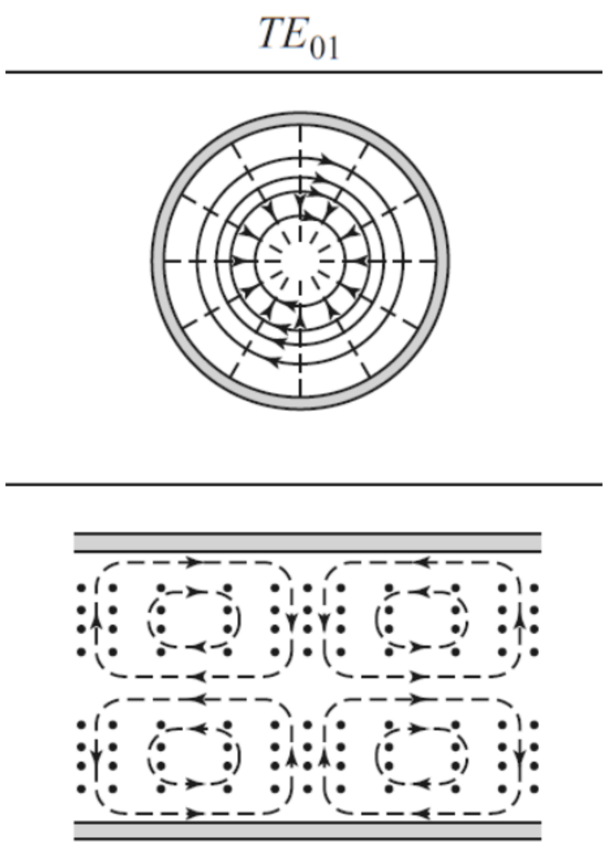
- Transmission lines help contain electromagnetic energy and transport it from one area to another
- Two common types used in the RF accelerator context:
  - Waveguides (Sami talked about them, so I assume you know everything about them)
  - Coaxial lines

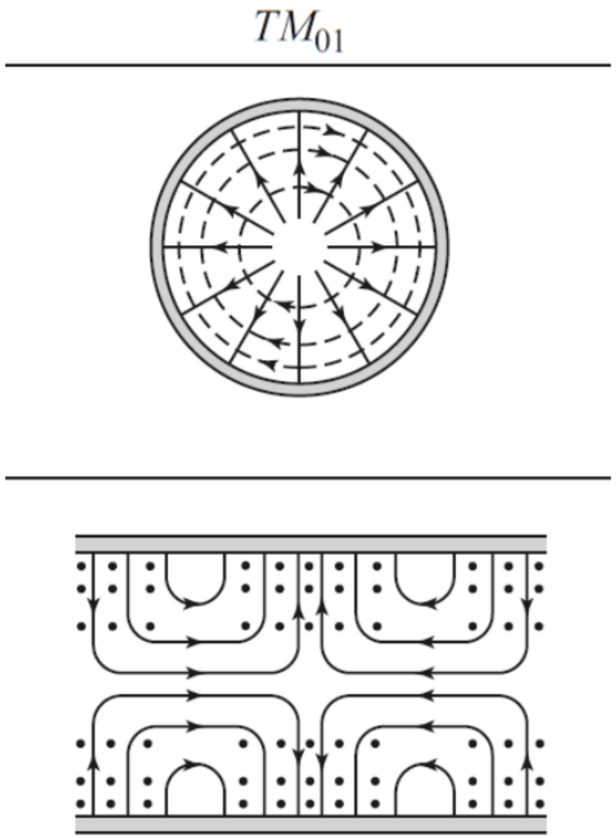




- Coaxial lines have no cutoff frequency, so they generally better for lower frequencies

Reminder on (some) Circular Waveguide Modes

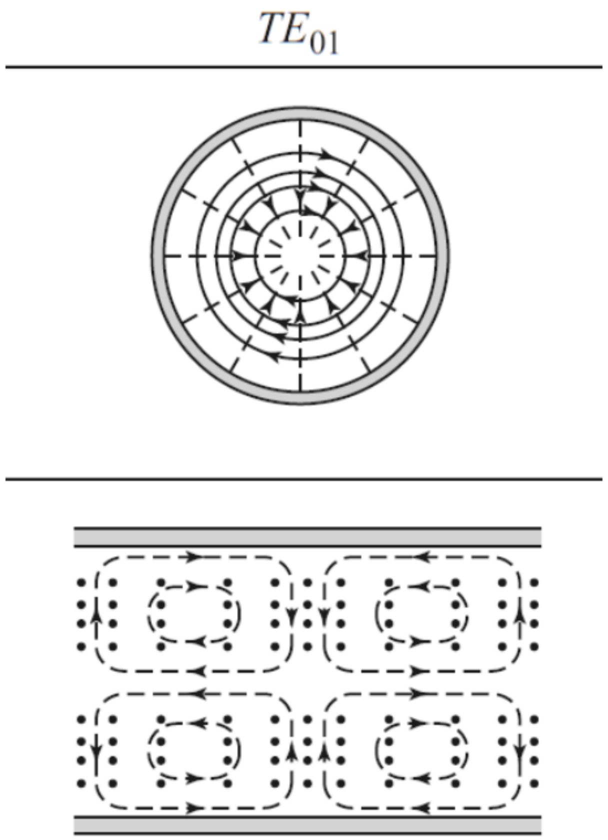


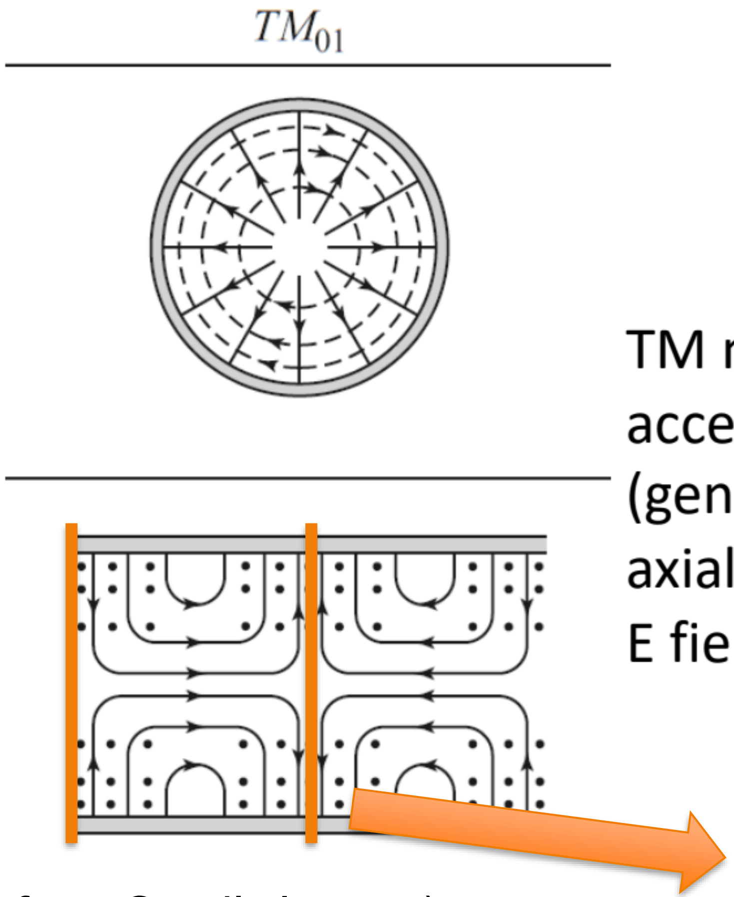


TM modes: our accelerating modes (generally), have axial component of E field

(stolen from Sami's lecture)

Reminder on (some) Circular Waveguide Modes





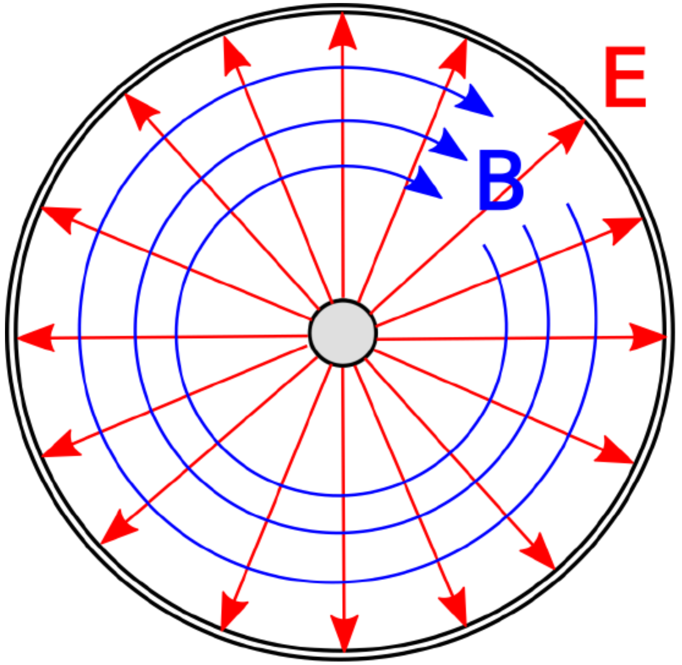
(stolen from Sami's lecture)

TM modes: our accelerating modes (generally), have axial component of E field

Add reflective boundaries!
This is the pillbox!

Coaxial Cables Support TEM Modes

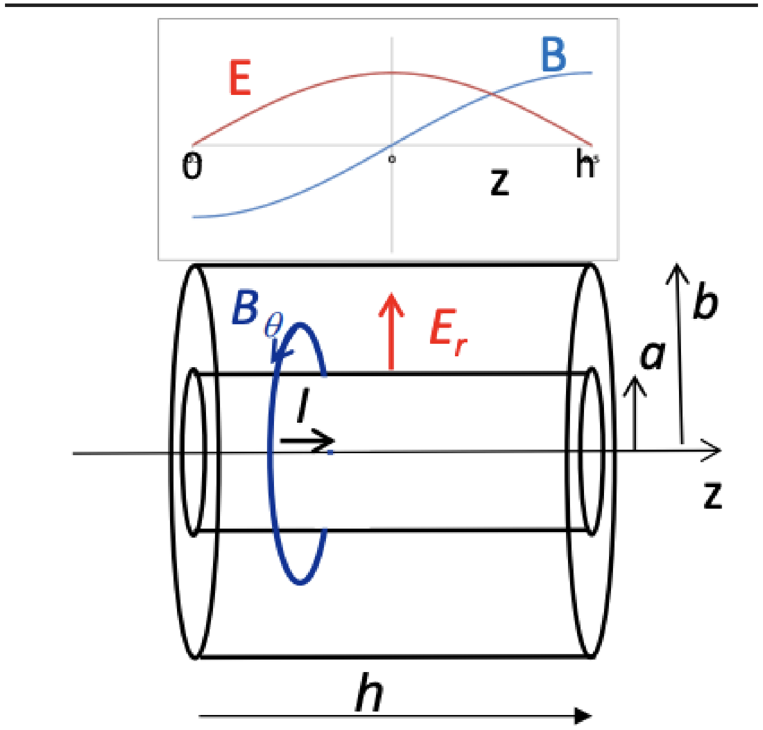
- Solving Maxwell’s equations for this geometry (two concentric cylinders) results in a **TEM (Transverse Electric Magnetic)** wave solution
  - Not supported in waveguides due to boundary conditions
  - E and B fields ”live” only in the transverse plane → no components along the direction of wave propagation



Wave propagation = into/out of page

Coaxial Cable to Coaxial (half-wave) Cavity

- Using same approach as circular waveguides for pillbox → apply reflective walls to "ends" of the coaxial line to create standing wave (a PEC or short BC at the cavity ends)



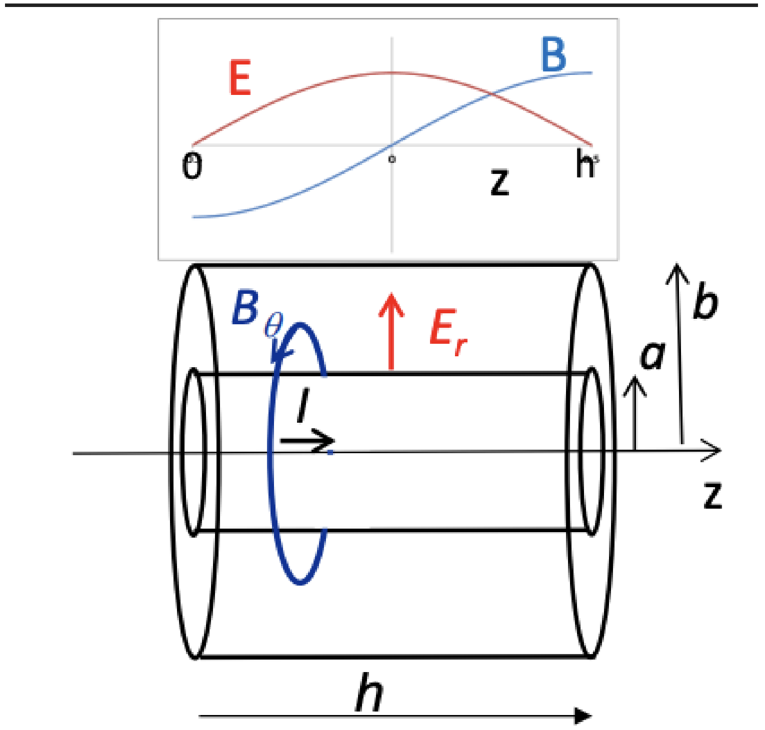
$$
\left.\begin{aligned}
B_\theta &= \frac{\mu_0 I_0}{\pi r}\cos\frac{p\pi z}{h}e^{j\omega t}\\
E_r &= -j\frac{\eta I_0}{\pi r}\sin\frac{p\pi z}{h}e^{j\omega t}
\end{aligned}\right\} \text{90 deg. out of phase}
$$

Cavity height sets the frequency: $h = \frac{\lambda}{2} \rightarrow \mathrm{f} = \frac{c}{2h}$

From Bob Laxdal's SRF25 tutorial slides: https://meow.elettra.eu/89/pdf/THTUT03_talk.pdf

Coaxial Cable to Coaxial (half-wave) Cavity

- Using same approach as circular waveguides for pillbox → apply reflective walls to "ends" of the coaxial line to create standing wave (a PEC or short BC at the cavity ends)



$$\left.\begin{aligned} B_\theta &= \frac{\mu_0 I_0}{\pi r}\cos\frac{p\pi z}{h}e^{j\omega t} \\ E_r &= -j\frac{\eta I_0}{\pi r}\sin\frac{p\pi z}{h}e^{j\omega t} \end{aligned}\right\} \text{90 deg. out of phase}$$

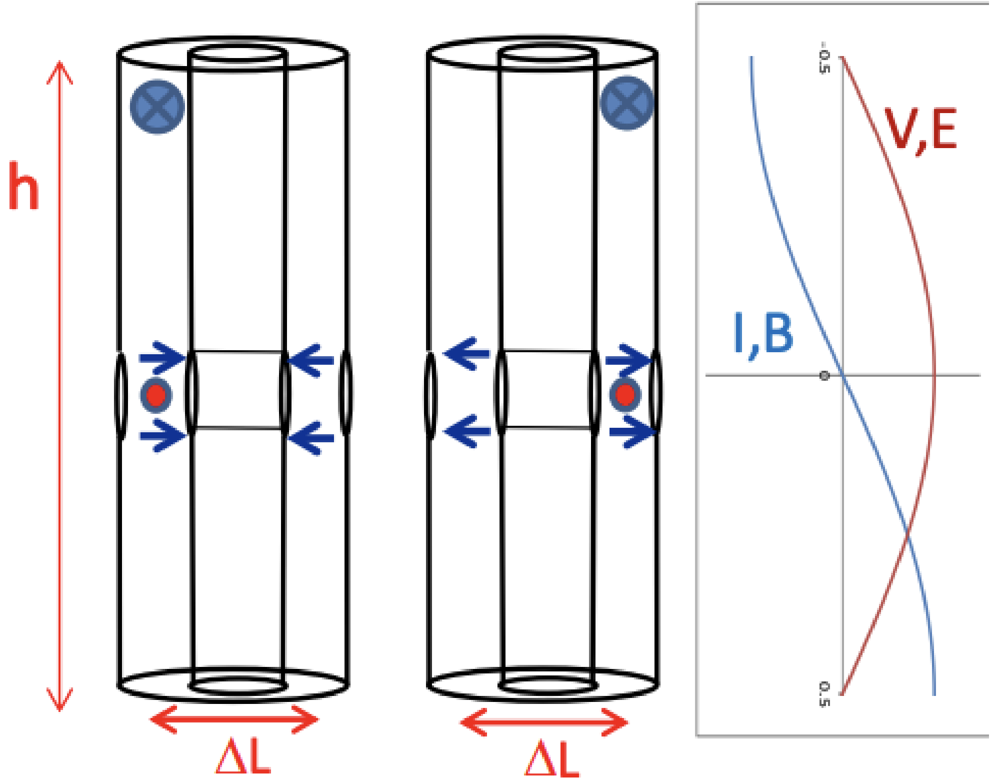
Cavity height sets the frequency: $h = \frac{\lambda}{2} \rightarrow \mathrm{f} = \frac{c}{2h}$

If I want to accelerate a beam with this geometry, where should I inject the particles?

From Bob Laxdal's SRF25 tutorial slides: https://meow.elettra.eu/89/pdf/THTUT03_talk.pdf

Acceleration with Coaxial (half-wave) Cavity

- Adding beam ports in the center of the cavity allows for a 2-gap acceleration structure
  - To achieve synchronism, the distance between gaps must allow the field to undergo a 180 deg. Phase shift



$$\Delta t = \frac{\Delta L}{v_0} = \frac{\Delta L}{\beta_0 c}$$

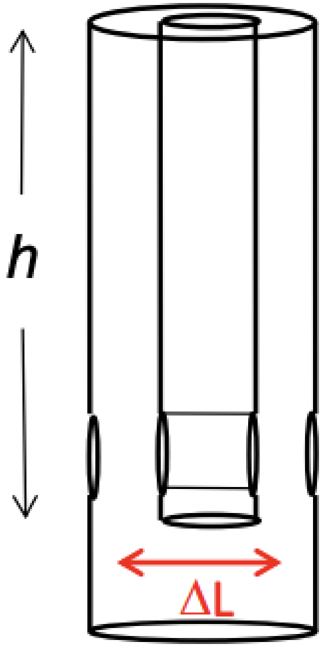
$$\Delta t = \frac{T}{2} = \frac{1}{2f} = \frac{\lambda}{2c}$$

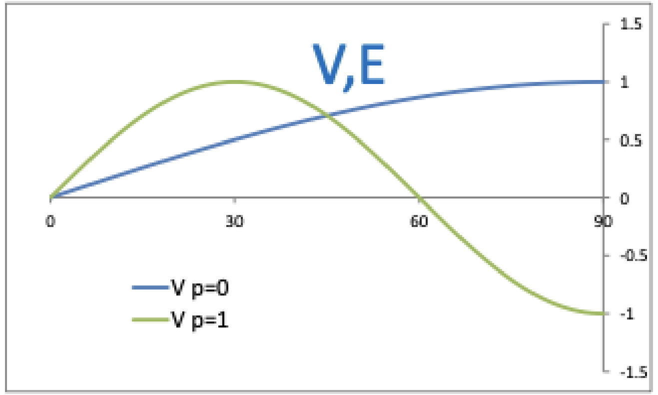
so $\Delta L = \beta_0 c \frac{\lambda}{2c} = \frac{\beta_0 \lambda}{2}$

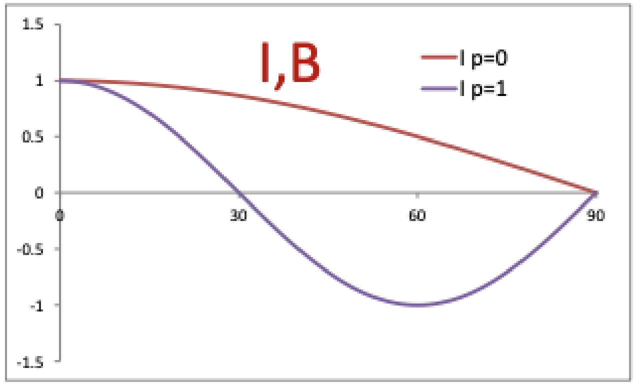
From Bob Laxdal's SRF25 tutorial slides: https://meow.elettra.eu/89/pdf/THTUT03_talk.pdf

Quarter-wave Coaxial Cavity

- For acceleration at even lower velocities (with lower frequencies), one of the BC's can be changed to "open" (PMC)
  - Like for HWR, $h = {}^{\lambda}/_{4}$ (height of IC) is set by the desired wavelength



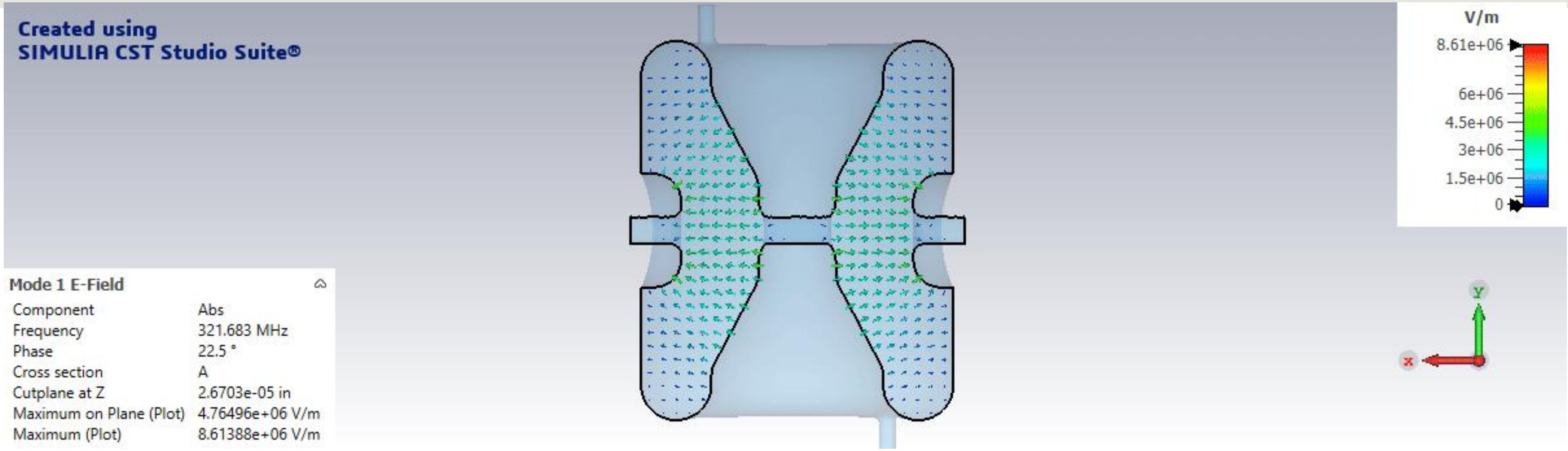




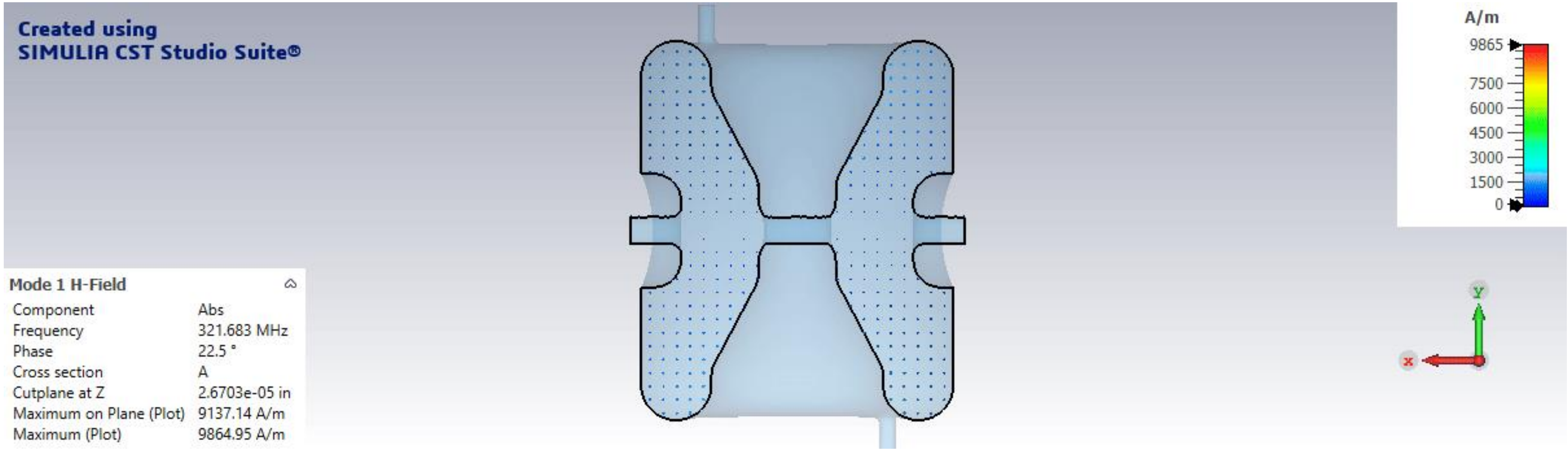
From Bob Laxdal's SRF25 tutorial slides: https://meow.elettra.eu/89/pdf/THTUT03_talk.pdf

Some HWR Field Maps (from CST MWS)

E-field

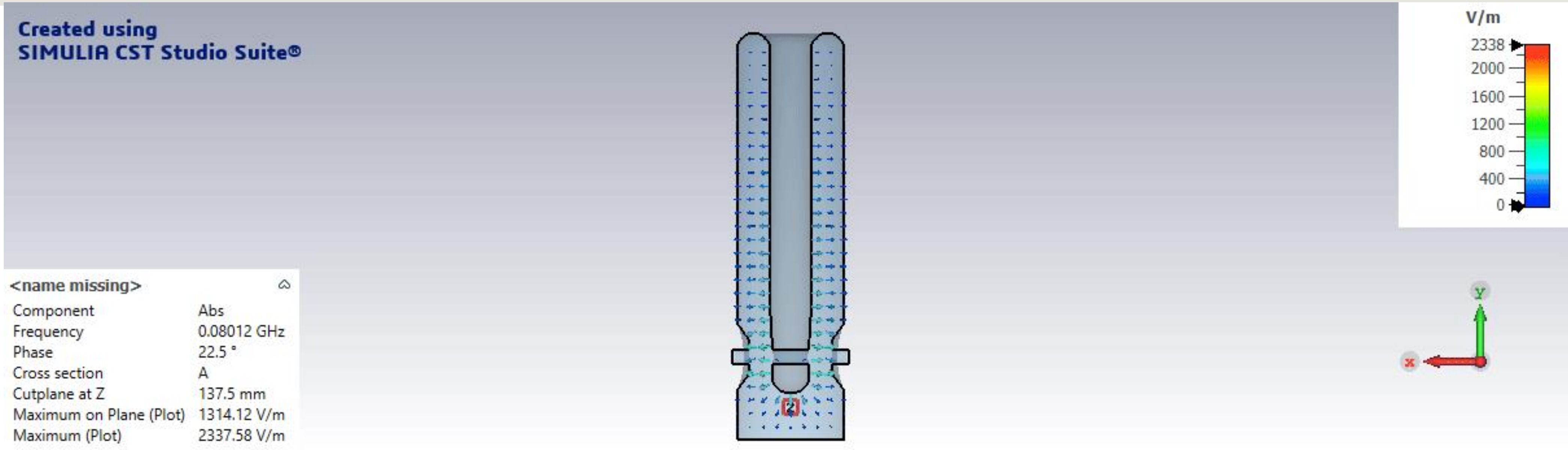


H-field

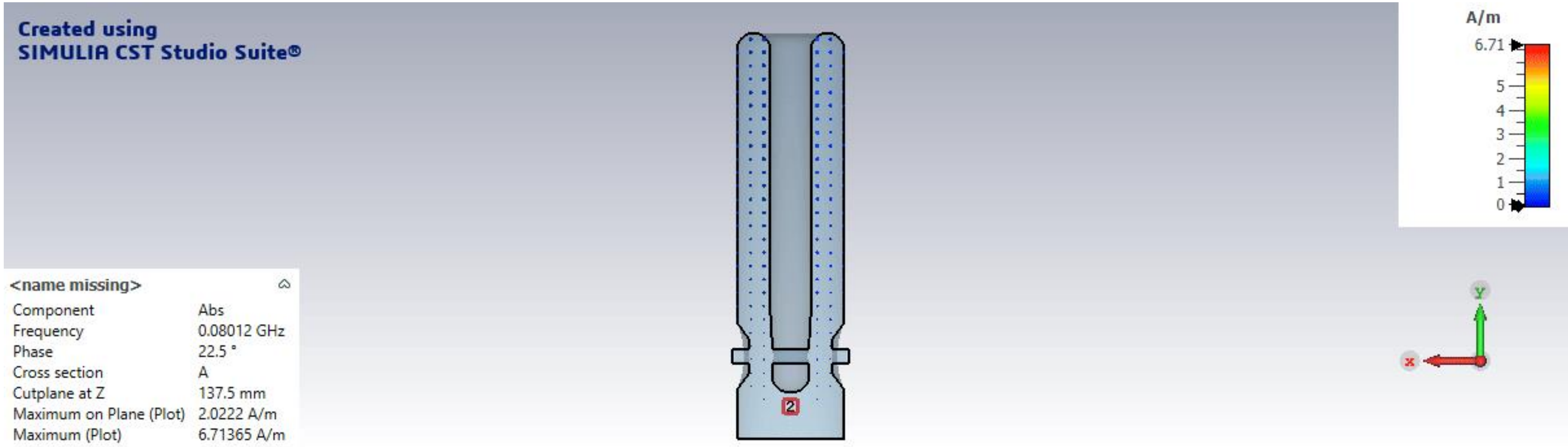


Some QWR Field Maps (from CST MWS)

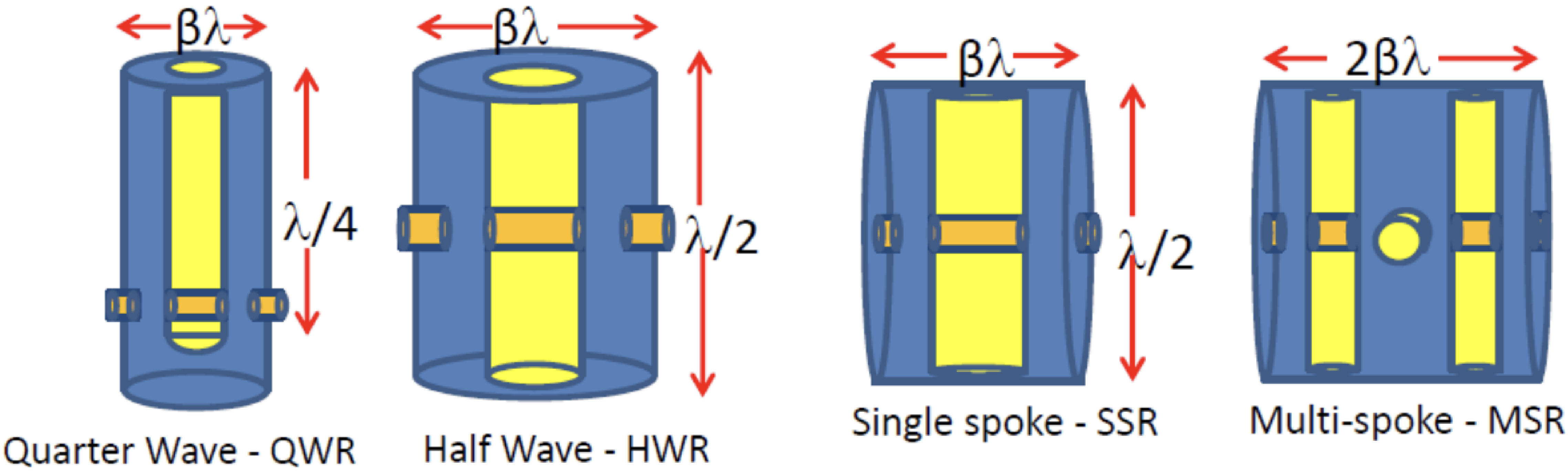
E-field



H-field

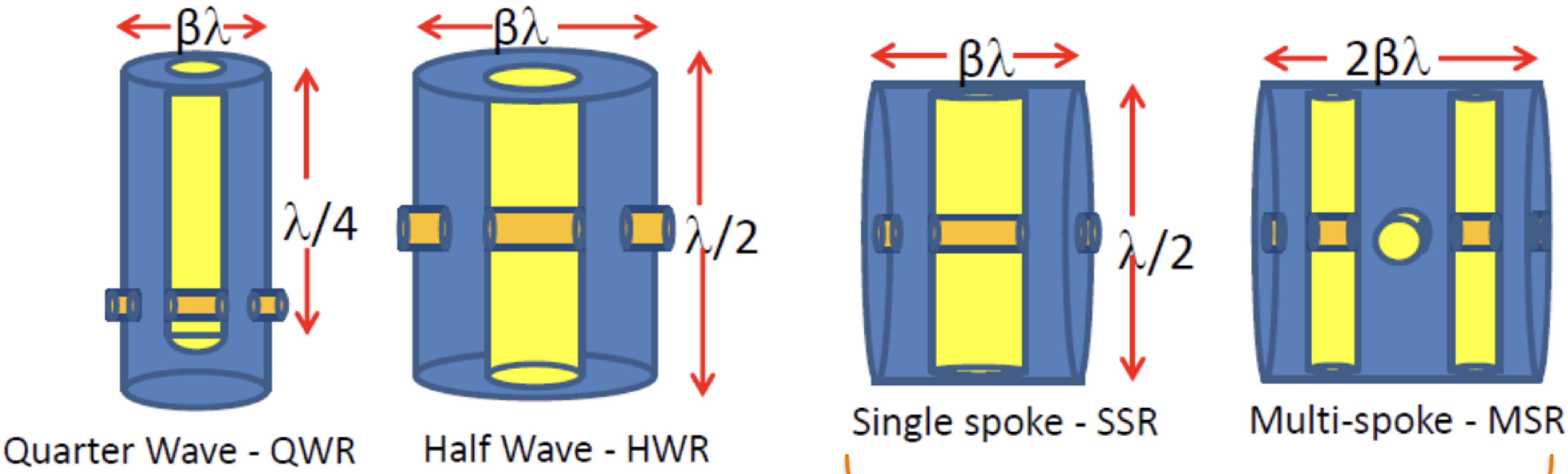


4 Main Types of Coaxial Resonators



From Bob Laxdal's SRF25 tutorial slides: https://meow.elettra.eu/89/pdf/THTUT03_talk.pdf

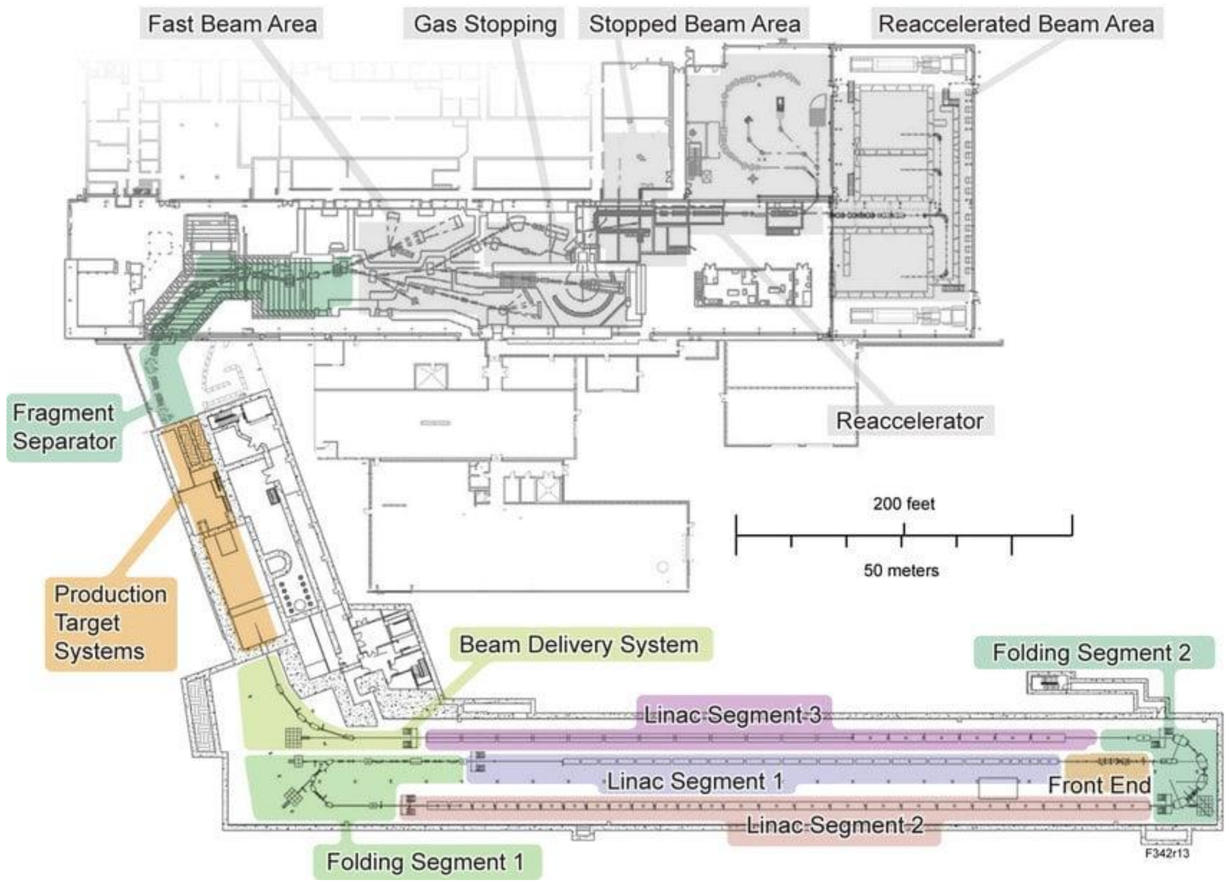
4 Main Types of Coaxial Resonators

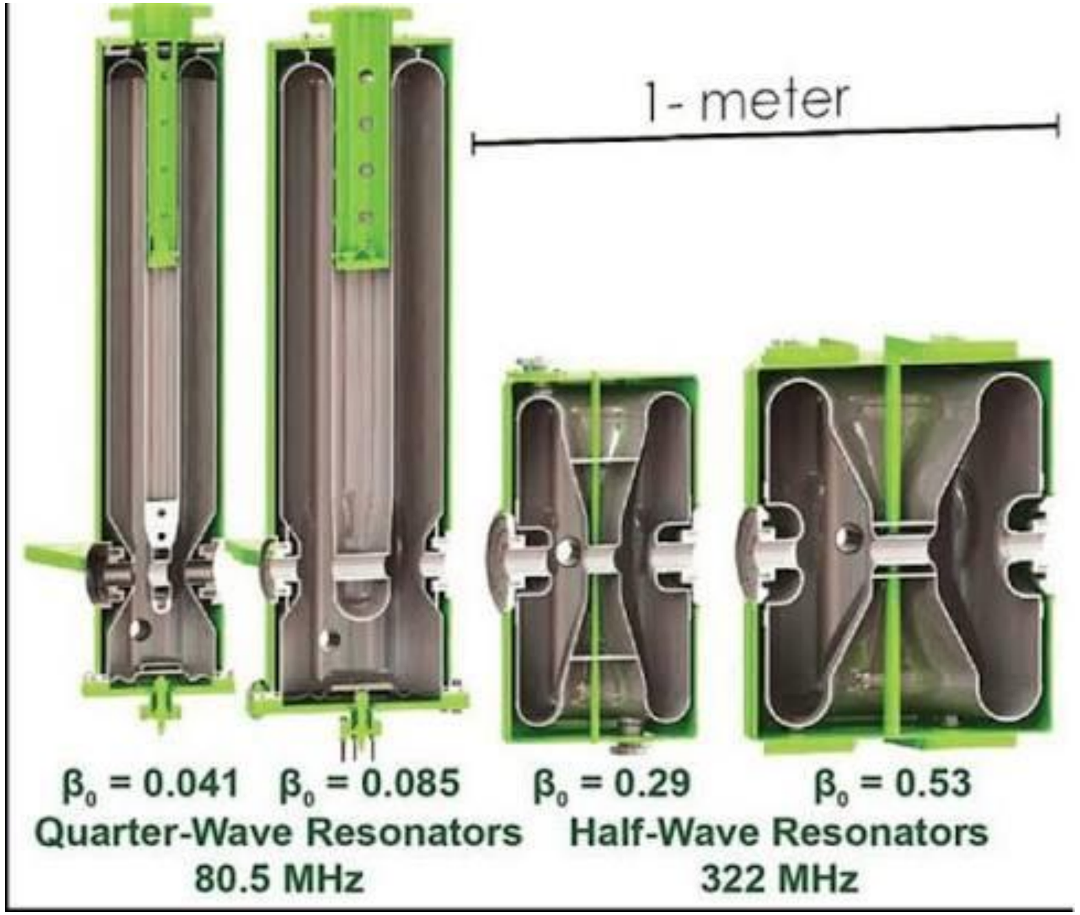


IC is rotated by 90 deg.

From Bob Laxdal's SRF25 tutorial slides: https://meow.elettra.eu/89/pdf/THTUT03_talk.pdf

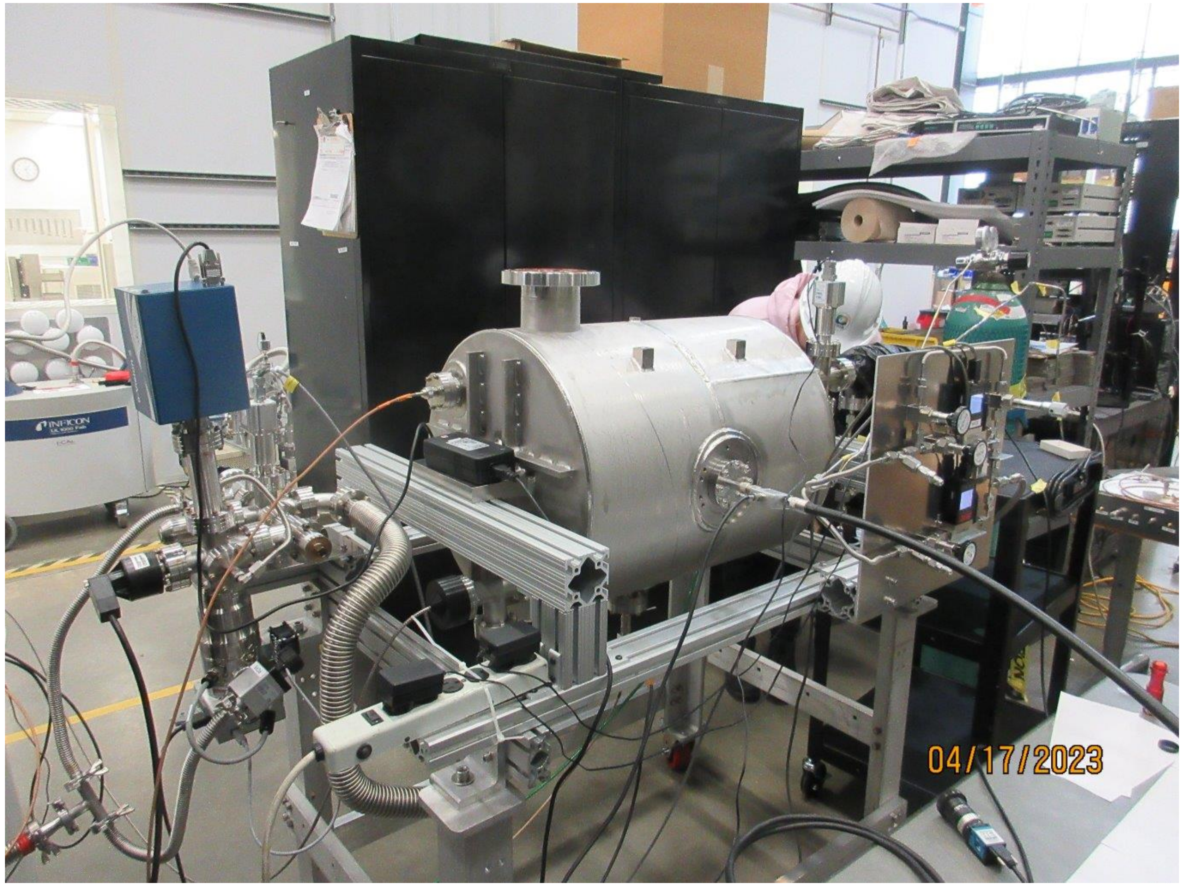
Some Coaxial Cavity Examples: FRIB at MSU





Some Coaxial Cavity Examples: FRIB at MSU

80.5 MHz QWR

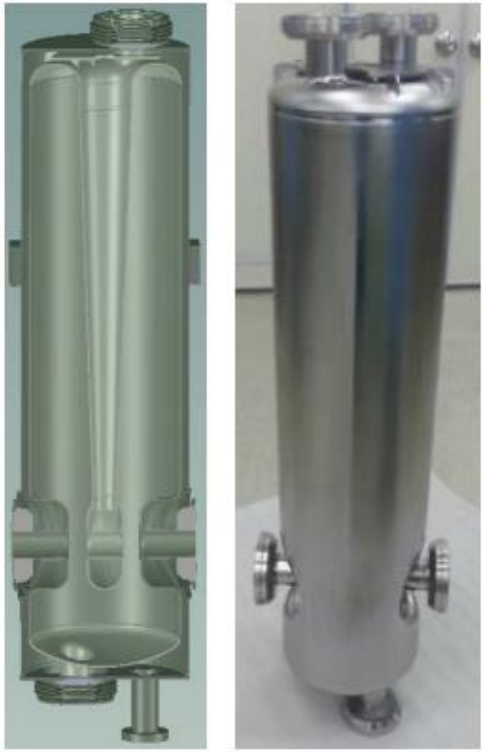


322 MHz HWR

Some Coaxial Cavity Examples: ANL, IJCLab, RAON

ANL: 87 MHz QWR for ATLAS

IJCLab: 88.05 MHz QWR for SPIRAL2



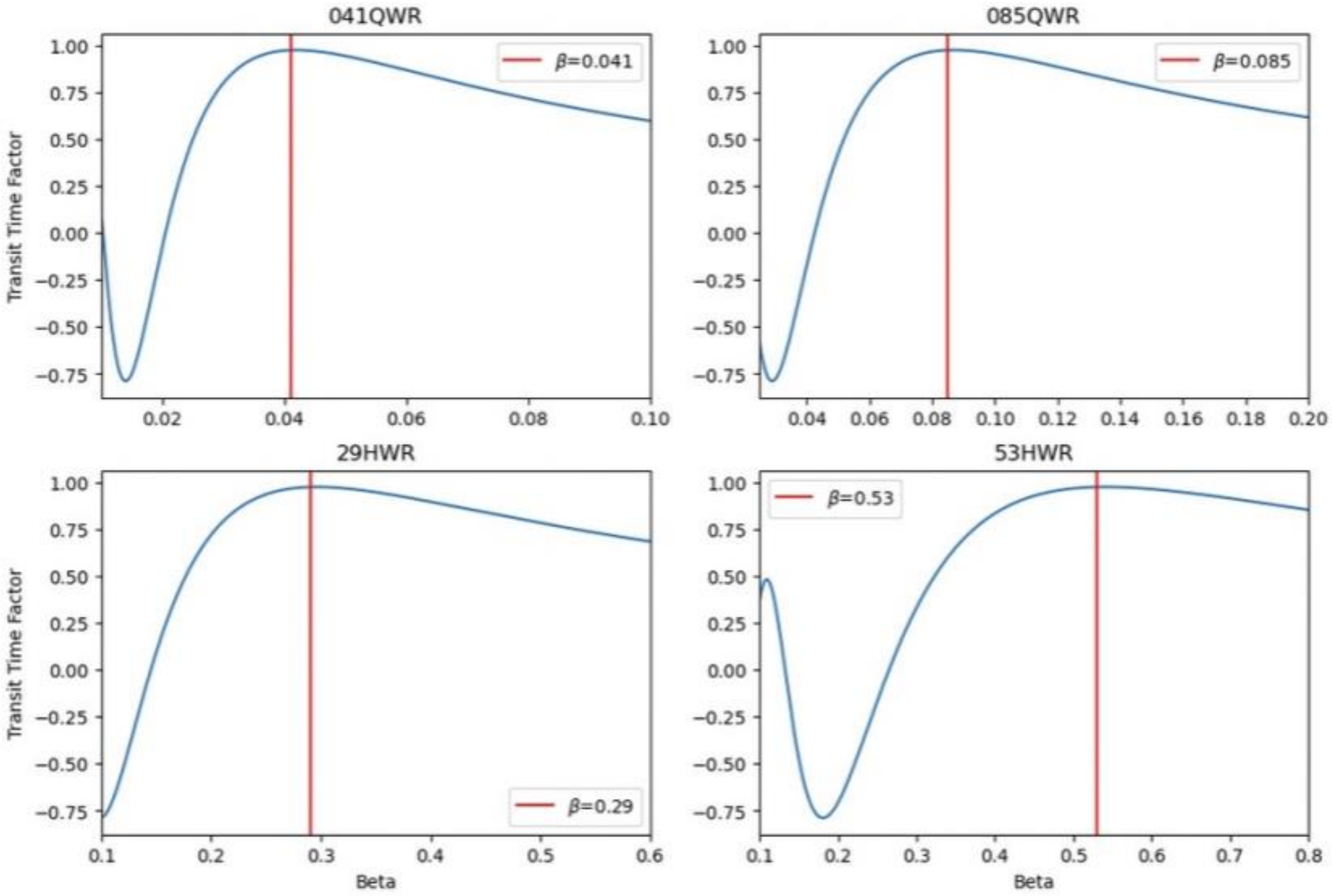
RAON β=0.047, 81.25 MHz

So what about the TTF for these 2-gap cavities?

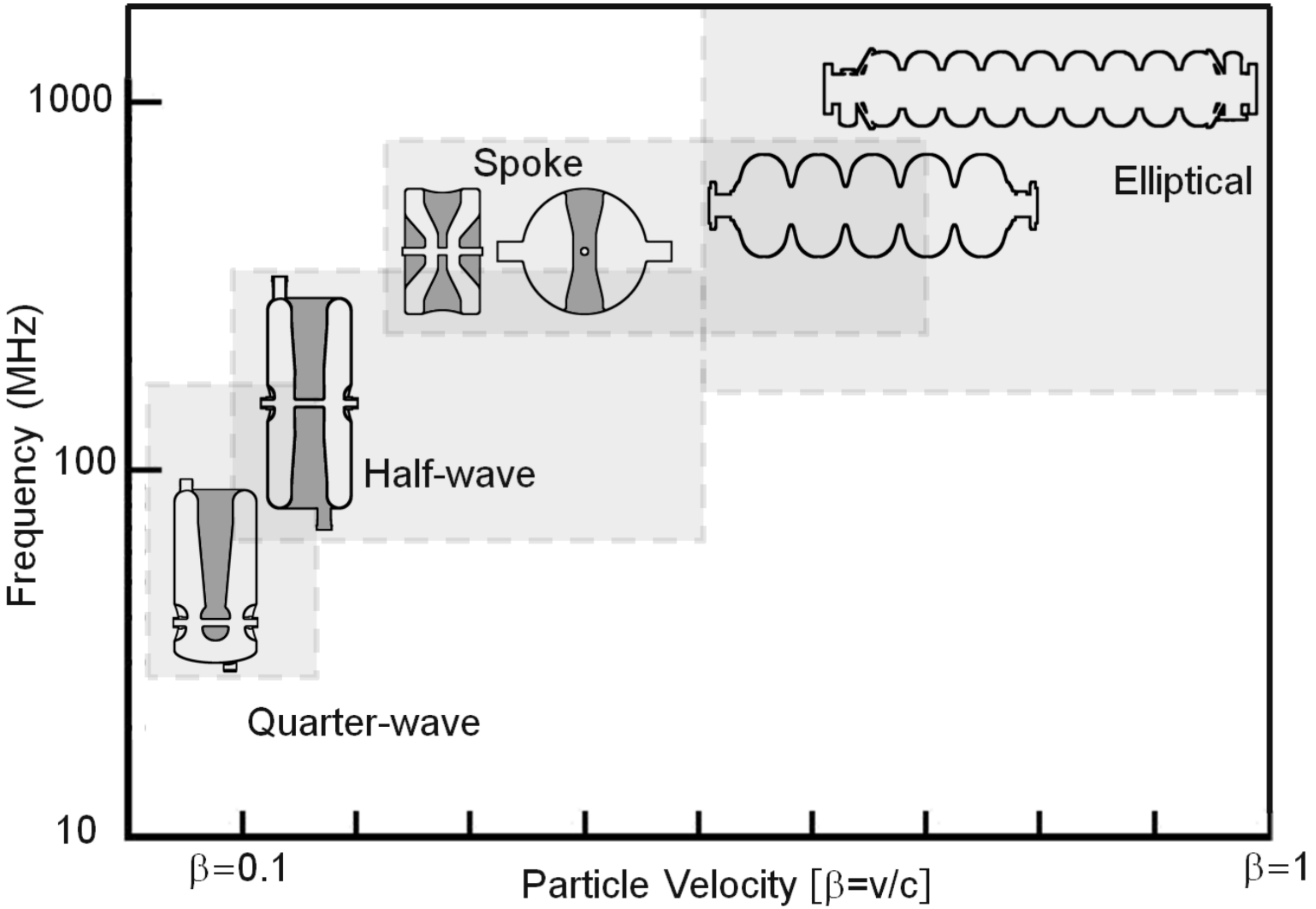
- Possible to derive a 2-gap TTF:

$$T = \frac{\sin\left({}^{\pi g}/_{\beta\lambda}\right)}{{}^{\pi g}/_{\beta\lambda}} * \sin\left({}^{\pi \beta_s}/_{2\beta}\right)$$

- Plotting TTF for each FRIB coaxial cavity



Cavity Types with Increasing Velocity



Source: https://indico.in2p3.fr/event/9782/contributions/50510/attachments/40912/50689/03_-_TEMcavityDesign_-_Kelly.pdf

Cavity Design + Optimization Steps (in general)

Cavity Design + Optimization Steps (in general)

- Pick your favorite EM simulation tool:
  - Commercial: Ansys HFSS, CST MWS, COMSOL
  - Specialized: ACE3P (from SLAC), SUPERFISH/Possion (from LANL)

Cavity Design + Optimization Steps (in general)

- Pick your favorite EM simulation tool:
  - Commercial: Ansys HFSS, CST MWS, COMSOL
  - Specialized: ACE3P (from SLAC), SUPERFISH/Possion (from LANL)
- Start with a baseline cavity model
- **1st Goal: optimize the RF parameters (EM design), BUT keep in mind:**
  - **Structure needs to be mechanically stable**
  - **Fabrication costs/abilities**

Reminder on Figures of Merit

FOM Name:	Unit:	Symbol:	Equation:
Frequency	Hz	f_0	--
Accelerating Field	MV/m	E_{acc}	$\frac{V_{acc}}{L_{cav}} = \frac{\left\lvert \int Ez\,(r=0,z)e^{ikz}dz \right\rvert}{L_{cav}}$
Transit Time Factor	--	T	$\frac{\left\lvert \int Ez\,(r=0,z)e^{ikz}dz \right\rvert}{\left\lvert \int Ez\,(r=0,z) \right\rvert}$
Shunt Impedance	Ohms	R_{sh}/Q	${V_{acc}}^2/\omega_0 U$
Quality Factor	--	Q_0	$\omega_0 U/P$
Stored Energy	J	U	$\frac{\varepsilon}{2}\int E^2 dV = \frac{\mu}{2}\int H^2 dV$
Geometry Factor	Ohms	G	$\omega\mu\frac{\int H^2\,dV}{\int H^2\,dS}$
E-field Peak / Accelerating Field	n/a	E_{peak}/E_{acc}	--
B-field Peak / Accelerating Field	mT/MV/m	B_{peak}/E_{acc}	--

During the design/optimization process, we want to:

FOM Name:	Unit:	Symbol:	Equation:	
Frequency	Hz	f_0	--	Keep constant!
Accelerating Field	MV/m	E_{acc}	$\frac{V_{acc}}{L_{cav}} = \frac{\left\lvert \int Ez\,(r=0,z)e^{ikz}dz \right\rvert}{L_{cav}}$	
Transit Time Factor	--	T	$\frac{\left\lvert \int Ez\,(r=0,z)e^{ikz}dz \right\rvert}{\left\lvert \int Ez\,(r=0,z) \right\rvert}$	↑ = more eff. acceleration
Shunt Impedance	Ohms	R_{sh}/Q	${V_{acc}}^2/\omega_0 U$	↑ = more eff. acceleration
Quality Factor	--	Q_0	$\omega_0 U/P$	
Stored Energy	J	U	$\frac{\varepsilon}{2}\int E^2 dV = \frac{\mu}{2}\int H^2 dV$	
Geometry Factor	Ohms	G	$\omega\mu\frac{\int H^2\,dV}{\int H^2\,dS}$	↑ = lower loss
E-field Peak / Accelerating Field	n/a	E_{peak}/E_{acc}	--	↓ = impacts field emission
B-field Peak / Accelerating Field	mT/MV/m	B_{peak}/E_{acc}	--	↓ = impacts quench field

Cavity Design + Optimization Steps (in general)

- Pick your favorite EM simulation tool:
  - Commercial: Ansys HFSS, CST MWS, COMSOL
  - Specialized: ACE3P (from SLAC), SUPERFISH/Possion (from LANL)
- Start with a baseline cavity model (for our case, the pillbox)
- **1st Goal: optimize the RF parameters (EM design), BUT keep in mind:**
  - **Structure needs to be mechanically stable**
  - **Fabrication costs/abilities**
- Shaping of the RF cavity:

1. Achieve the desired f_0 and β_0
2. Try to maximize G, R/Q, while minimizing peak fields **AND** keeping f_0 and β_0 constant
3. Check for multipacting barriers (some EM simulation tools can do this) → want to eliminate barriers near the operating gradients

Cavity Design + Optimization Steps (in general)

- Pick your favorite EM simulation tool:
  - Commercial: Ansys HFSS, CST MWS, COMSOL
  - Specialized: ACE3P (from SLAC), SUPERFISH/Possion (from LANL)
- Start with a baseline cavity model (for our case, the pillbox)
- **1st Goal: optimize the RF parameters (EM design), BUT keep in mind:**
  - **Structure needs to be mechanically stable**
  - **Fabrication costs/abilities**
- Shaping of the RF cavity:
  1. Achieve the desired f_0 and β_0
  2. Try to maximize G, R/Q, while minimizing peak fields **AND** keeping f_0 and β_0 constant
  3. Check for multipacting barriers (some EM simulation tools can do this) → want to eliminate barriers near the operating gradients
- **2nd Goal: If relevant for the application, can simulate wake potentials (CST, ECHO, GdfidL)**

Cavity Design + Optimization Steps (in general)

- Pick your favorite EM simulation tool:
  - Commercial: Ansys HFSS, CST MWS, COMSOL
  - Specialized: ACE3P (from SLAC), SUPERFISH/Possion (from LANL)
- Start with a baseline cavity model (for our case, the pillbox)
- **1st Goal: optimize the RF parameters (EM design), BUT keep in mind:**
  - **Structure needs to be mechanically stable**
  - **Fabrication costs/abilities**
- Shaping of the RF cavity:
  1. Achieve the desired f_0 and β_0
  2. Try to maximize G, R/Q, while minimizing peak fields **AND** keeping f_0 and β_0 constant
  3. Check for multipacting barriers (some EM simulation tools can do this) → want to eliminate barriers near the operating gradients
- **2nd Goal: If relevant for the application, can simulate wake potentials (CST, ECHO, GdfidL)**
- **3rd Goal: Thermal + Mechanical simulations**
  - Assess mechanical modes/frequencies + Lorentz force detuning (relevant for pulsed operation) + temperature profiles

Cavity Design + Optimization Steps (in general)

- Pick your favorite EM simulation tool:
  - Commercial: Ansys HFSS, CST MWS, COMSOL
  - Specialized: ACE3P (from SLAC), SUPERFISH/Possion (from LANL)
- Start with a baseline cavity model (fo
- **1st Goal: optimize the RF**
  - **Structure needs to be**
  - **Fabrication costs/al**
- Shaping of the RF ca
  1. Achieve the desired
  2. Try to maximize G, R/ ... ant
  3. Check for multipacting barr ... t to eliminate barriers near the operating gradients
- **2nd Goal: If relevant for the application, can simulate wake potentials (CST, ECHO, GdfidL)**
- **3rd Goal: Thermal + Mechanical simulations**
  - Assess mechanical modes/ frequencies + Lorentz force detuning (relevant for pulsed operation) + temperature profiles

Cavity design is an iterative process. Once the design is changed to assess a problem (multipacting barrier, addressing a mechanical mode, etc), return to the RF parameter optimization and repeat.

Cavity/Cryomodule Design Flowchart

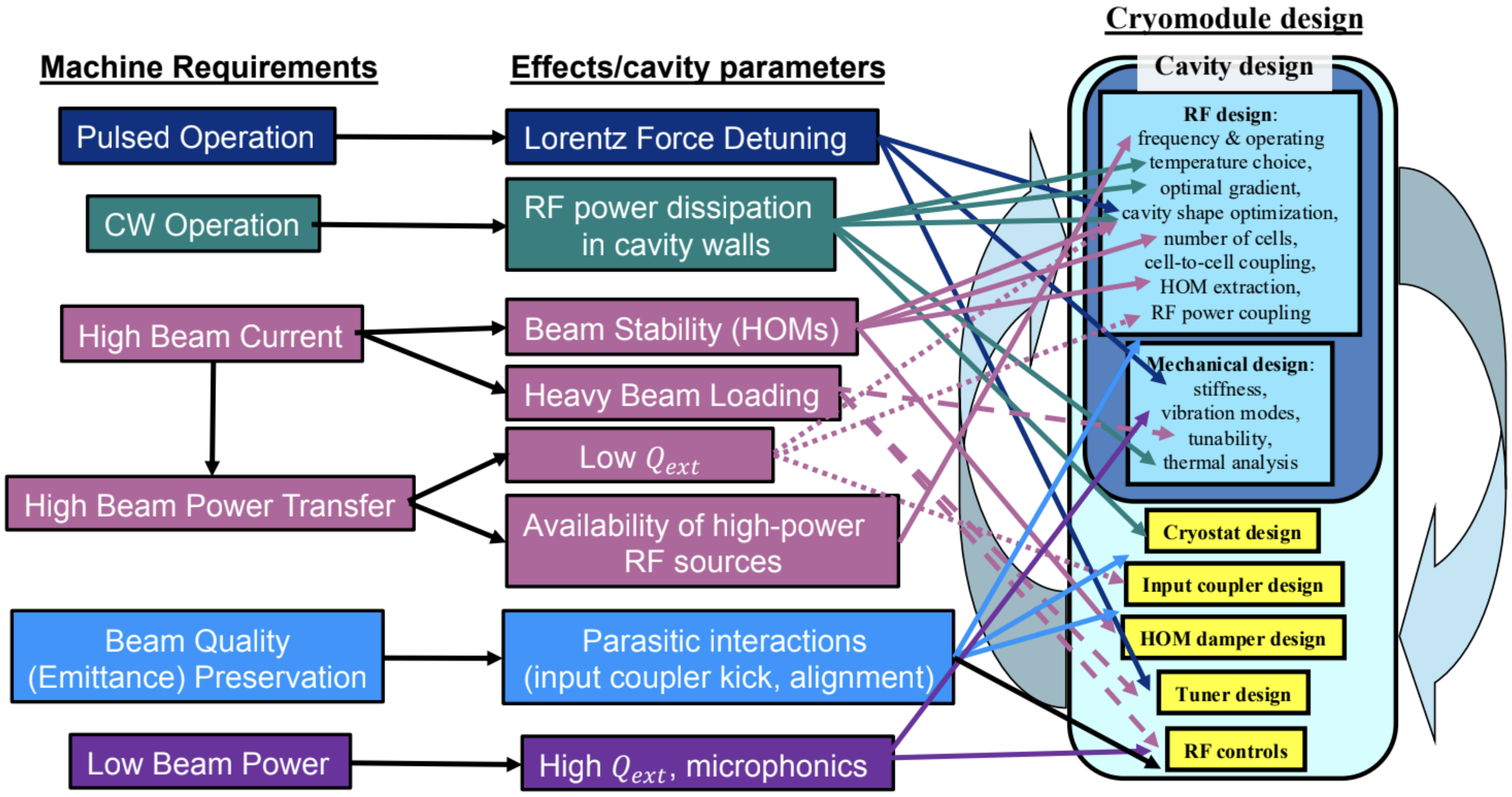
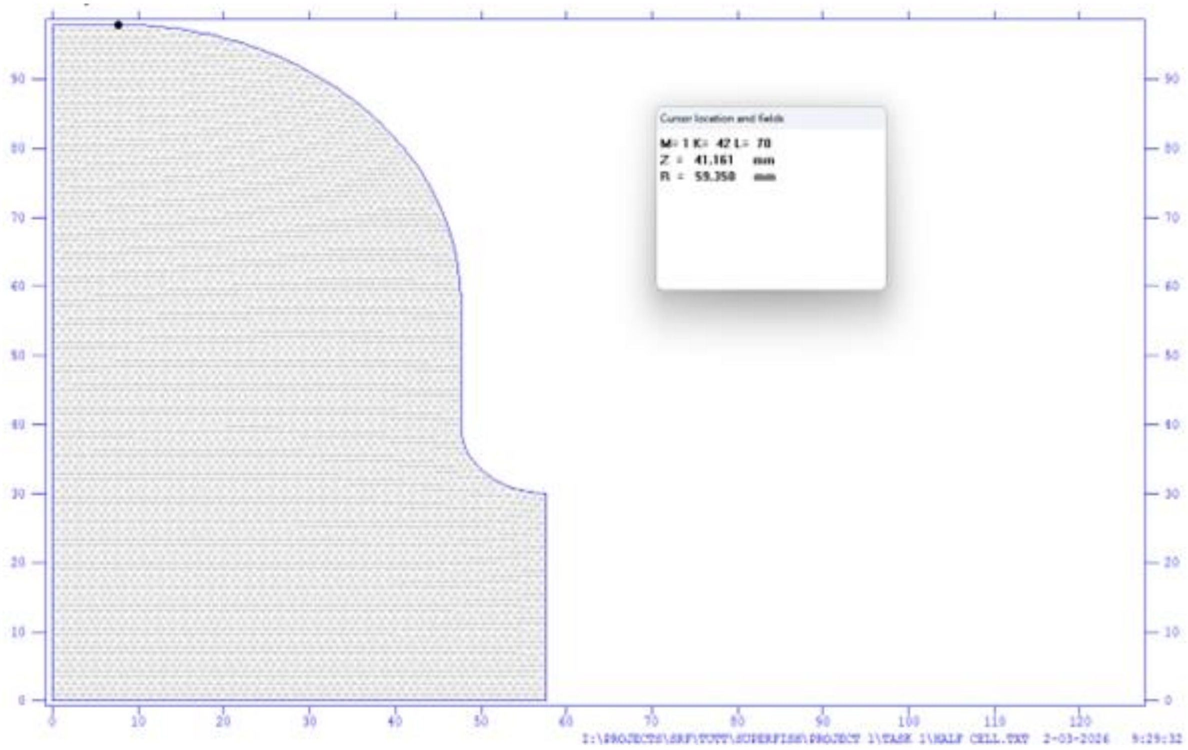


Image from previous USPAS lecture (11): https://sites.google.com/view/uspas2025-rf-superconductivity/home

Single-Cell Optimization with SUPERFISH (1.3 GHz)

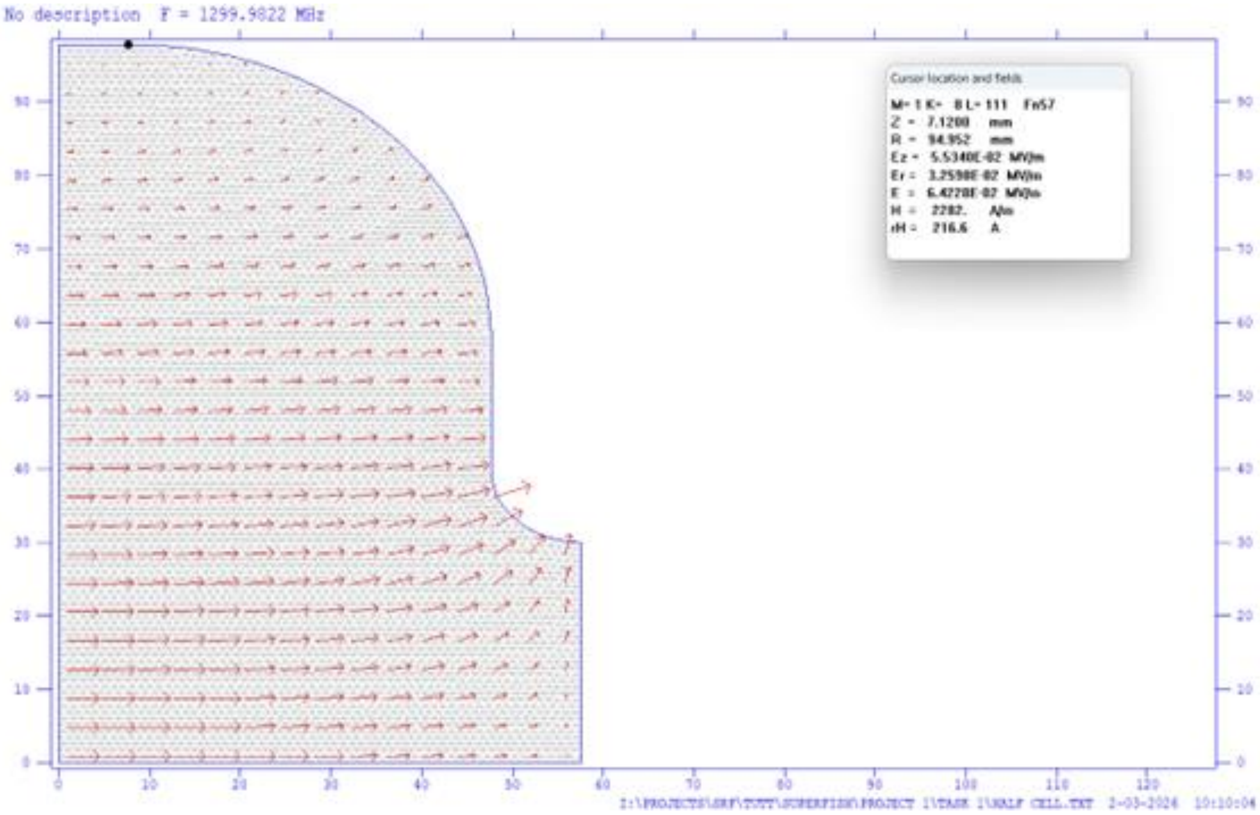
- Superfish as a FEM solver that can find RF EM fields in 2D or axially symmetric cases (like a single cell/pillbox cavity)
- Need to specific points/curves in an input file along with boundary conditions:



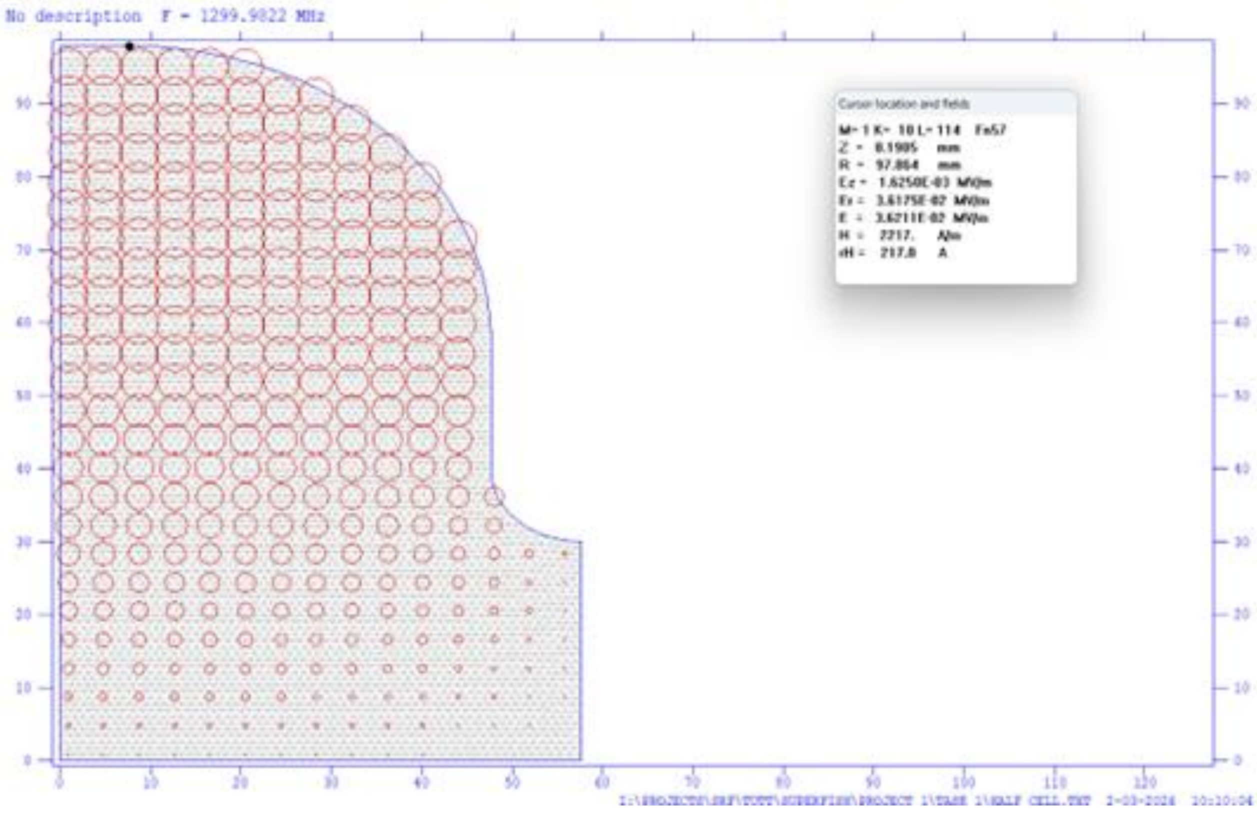

```
&REG
; Basics
KPROB=1 ; Superfish
ICYLIN=1 ; Cylindrical coords X -> z, Y -> r
; Boundary conditions for enclosing rectangle
NBSUP=1 ; BC for top edge: metal
NBSLO=0 ; BC for axis
NBSRT=0 ; BC for iris plane
NBSLF=1 ; BC for equator plane
; Start freq, particle beta
FREQ=1300.0 ; starting frequency in MHz
KMETHOD=1; use BETA to compute transit time
BETA=1.0 ; v/c
;
SCCAV=1 ;
; Material properties
MAT=1 ; Vacuum in cavity volume
IRTYPE=1 ; Superconductor formula for Rs
TEMPK=2.0; operating temperature (K)
TC=9.2 ; Tc of metal (K)
RESIDR=2.0E-9 ; residual resistance (Ohms)
; Drive point
XDRI=7.65 ; drive pt x
YDRI=97.92 ; drive pt y
; Units and mesh size
CONV=0.1 ; coords in mm
DX=1.0 ; mesh spacing
&
;Half-cell shape
&PO X=0.0, Y=0.0 &
&PO X=0.0, Y=97.92 &
&PO X=7.65, Y=97.92 &
&PO NT=2, X0=7.65, Y0=57.92, R=40.0, THETA=0.0 &
&PO X=47.65, Y=40.0 &
&PO NT=2, X0=57.65, Y0=40.0, R=10.0, THETA=-90.0 &
&PO X=57.65, Y=0.0 &
&PO X=0.0, Y=0.0 &
```

E/H Fields for Baseline 1.3 GHz Single-Cell

E-field



H-field



FOM for Baseline 1.3 GHz Single-Cell

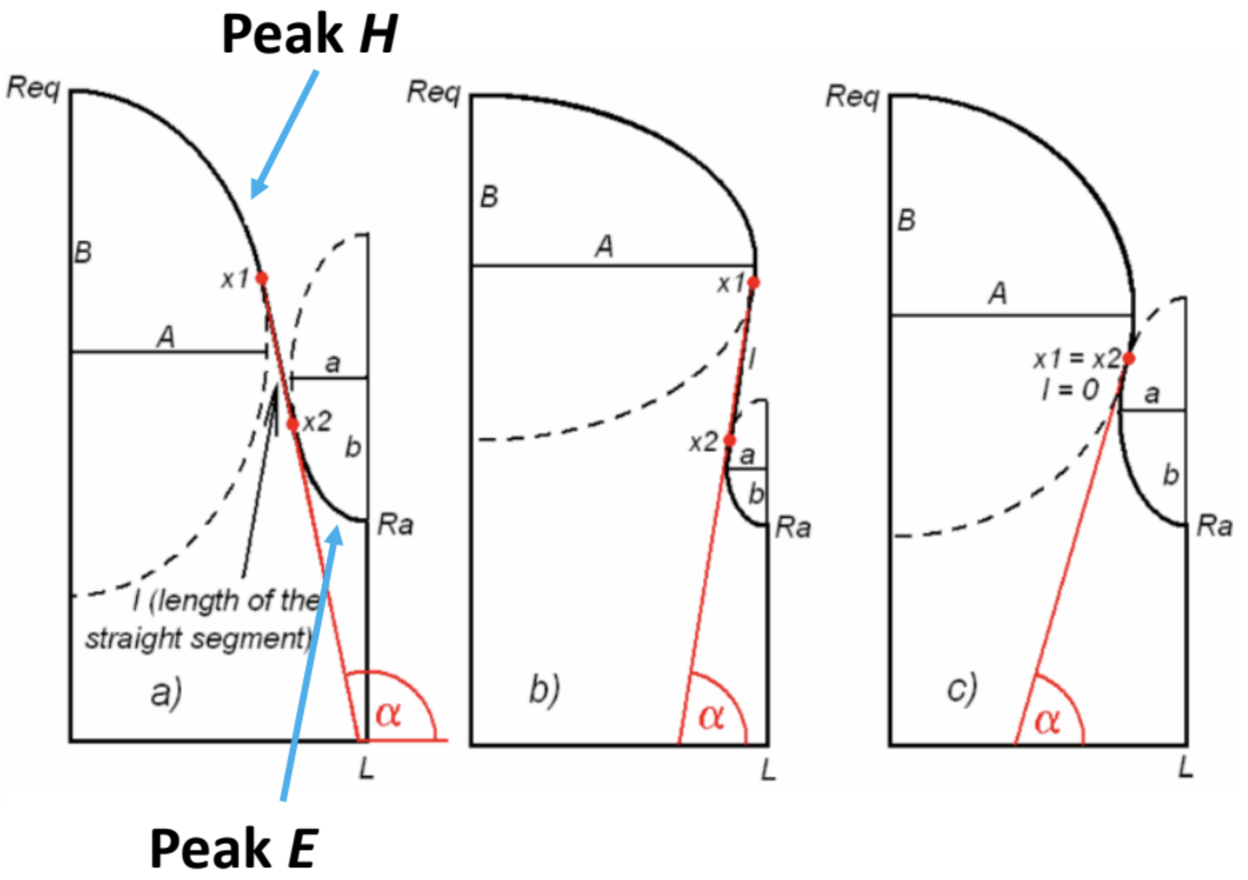
- Once the mesh is generated and fields are calculated → post-processing program (SFO) can be used to find the common FOM (but you should always check them for yourself)

FOM	Value	Scaled for Full Cell	Unit
f_0	1299.982	-	MHz
β	1.0	-	-
T	0.761	-	-
E_{acc}	0.761	-	MV/m
U	3.4856	6.9712	mJ
R_s	18.7956	-	nOhm
P_{walls}	1.936	3.872	mW
Q	$1.470 * 10^{10}$	-	-
G	276.353	-	Ohm
R/Q	67.577	135.154	Ohm
E_{peak}/E_{acc}	2.2502	-	-
B_{peak}/E_{acc}	3.6767	-	mT/MV/m

Make sure to think about what values scale from a half-cell to a full-cell!

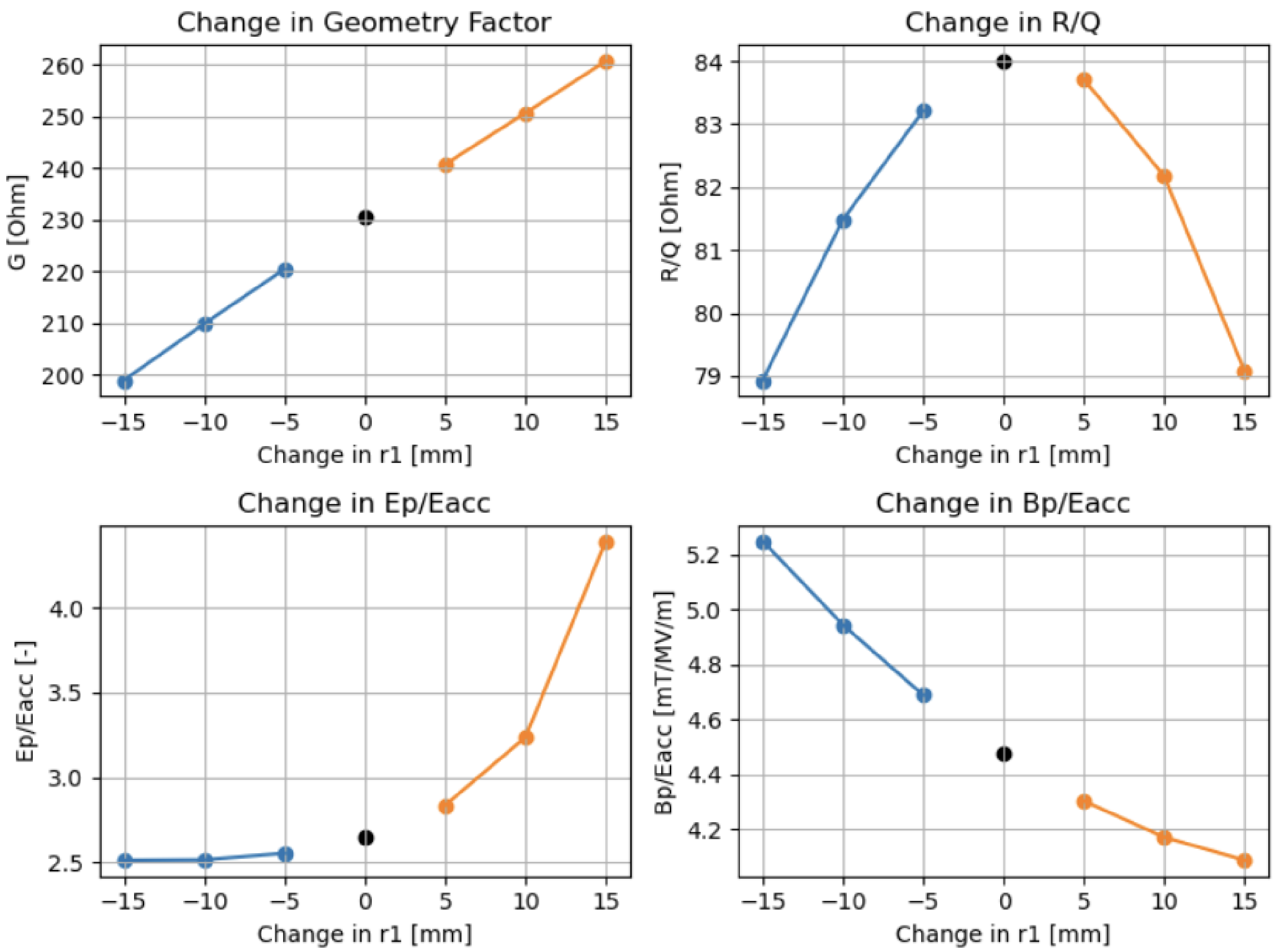
Time to optimize!

- Keep in mind:
  - Increase: R/Q and G
  - Reduce: Peak fields
  - Constant: frequency
- Typical approach:
  - Parametrize the shape such that you can make form constraint equations that keep things like the frequency (or effective length) constant
- One example:
  - Vary the radius of curvature of the top ellipse
  - Vary the radius of curvature of the bottom ellipse to offset the change in the frequency



(stolen from Sami's lecture, see her slides for link)

Sweep R1 (and correcting with R2)



Thanks! Questions?

- Interested in other cavity types and how they work? Look here:
  - USPAS class on SRF: https://sites.google.com/view/uspas2025-rf-superconductivity/home
    - » SRF fundamentals
    - » Beam-cavity interaction
    - » Crab cavities
    - » Cavity fabrication/design
    - » And more!
  - SRF tutorial sessions: https://meow.elettra.eu/89/session/1374-sat/index.html
    - » TM cavities
    - » TEM cavities
    - » RF controls
    - » Cryogenics
    - » High-power couplers